

Question Bank
Cloud Computing and Security (BIS613D)
(Modules 3, 4 and 5)

Module – 3

1. Compare and analyze the features and services of Google App Engine (GAE), IBM, Amazon Web Services (AWS), and Microsoft Azure.
2. Compare and analyze the features and services of Amazon Web Services (AWS) and Microsoft Azure in contrast with their services.
3. Describe the fat-tree topology and explain its use in data center network architecture.
4. Explain the five key requirements for an efficient cloud data center interconnection network.
5. What is a Warehouse-Scale Data Center? Explain its architecture.
6. Explain in detail about Warehouse Scale Computers with architecture.

Module – 4

10. What is a Trusted Hypervisor? Explain the major security challenges faced by mobile devices in cloud environments.
11. Discuss the security risks posed by shared VM images and Management Operating Systems in cloud computing.
12. Describe the hidden risks in cloud computing and analyze their impact on organizations.
13. Analyze the advantages of cloud technology for social networking applications.
14. Explain the key features and cloud services provided by Google in cloud computing environments.
15. Analyse the differences between Google File System (GFS), BigTable, and Chubby Lock Services.
16. Illustrate the application classification for parallel and distributed systems.
17. Demonstrate an Eucalyptus architecture for VM image management.

Module – 5

19. What are the benefits and challenges of using serverless computing in cloud environments?
20. What is Infrastructure as Code (IaC)? Explain how it is used in cloud environments.

21. Explain the concept of emerging cloud environments.
22. Illustrate OpenNebula architecture and its main components.
23. Explain the Sector/Sphere system architecture and Aneka architecture diagram.
24. Discuss the four emerging virtual infrastructure centers with a neat diagram.
25. Explain cloud resource deployment using an InterGrid Gateway (IGG) to allocate VMs from a local cluster to interact with the InterGrid Fabric (IFF) of a public cloud provider.